

Daniel Williford

Applied AI Systems Engineer

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PROFESSIONAL SUMMARY

Applied AI systems engineer with enterprise IT roots and 9 years building independent finance research, data, dashboard, and automation systems. I design and operate human reviewed AI services for workflows where evidence, provenance, governance, and reliability matter. Strong fit for applied AI, automation specialist, forward deployed engineering, AI infrastructure, AI solutions, and CloudOps/DevOps roles.

CORE SKILLS

Forward deployed delivery: stakeholder discovery, workflow mapping, success metrics, technical feasibility, acceptance criteria, rollout risks, runbooks, and operational handoff artifacts.

Automation and workflow systems: service boundaries, typed workflow contracts, APIs, dashboards, approval gates, exception handling, audit traces, and reusable workflow assets.

Agentic AI systems: tool using agents, role scoped context, RAG, vector retrieval, LangGraph workflows, eval gates, trace artifacts, human review, and deterministic task routing.

Backend and dashboards: Python, FastAPI, Pydantic, React/TanStack dashboards, CLI automation, REST APIs, validation services, containerized local demos, pytest validation, and hosted AI API integration.

Data and research systems: PostgreSQL, SQLite, PyArrow, Parquet, embeddings, vector search, source ledgers, data cleaning, freshness checks, schema validation, reproducible artifacts.

AI infrastructure operations: Linux services, local LLM endpoints, hosted API fallback, small model worker routing, systemd, journald, health checks, controlled restarts, VPN secured access, GPU workload awareness, and deployment ready service packaging.

Safety and observability: allowlisted tools, dry run defaults, path boundary checks, audit traces, approval records, logs, eval regressions, telemetry, production mutation guards.

PROFESSIONAL EXPERIENCE

Independent Automation, Finance Research, and AI Systems Developer | 2017 Present

- Built and operated internal automation systems for finance research, data ingestion, data cleaning, backtesting, dashboards, reporting, alerts, and review workflows.
- Developed human in the loop research tools where AI assists with evidence gathering, summarization, candidate ranking, paper trial monitoring, plots, and inspection packets while final decisions stay human owned.
- Maintained a small VPN secured services network supporting several users and multiple machines, including local AI services, bounded tool interfaces, file services, dashboards, and controlled service lifecycle management.
- Added validation discipline through source freshness checks, leakage checks, schema validation, eval harnesses, trace artifacts, and explicit execution boundaries.
- Operated cost aware local infrastructure for LLMs, analytics, dashboards, research automation, and GPU workloads.

Independent Video Production and Client Services | 2013 2020

- Ran client facing production work with scheduling, technical setup, media workflows, delivery requirements, customer communication, vendor coordination, and deadline driven execution.
- Transitioned full time into finance research systems and automation after 2020 as finance and software automation became the primary business focus.

IT Specialist, CSC / Raytheon | 2011 2013

- Supported enterprise IT across offices, labs, and restricted access environments.
- Performed installs, upgrades, break/fix troubleshooting, access checks, and network connectivity triage.
- Followed standardized configuration and validation procedures while coordinating with Linux administrators and network teams.

IT Technician, Unified School District 453 | 2009 2011

- Installed and maintained end user systems, supported helpdesk workflows, handled basic network troubleshooting, and improved documentation for repeat issues.

SELECTED SYSTEMS EVIDENCE

Human in the Loop ML Finance Research Copilot

- Built advisory research dashboards and agent workflow support for market and research feed ingestion, ML validation, backtest review, source freshness, paper trial monitoring, candidate ranking, plots, and human promotion decisions.

Agentic Workflow Ops Service

- Built local FastAPI + LangGraph workflow service for claims/policy style workflows with typed contracts, retrieval, traces, artifacts, approval controls, dry run behavior, source backed packets, and regression tests.

Agent Memory and Routing Harnesses

- Built role profiles, memory policy, allowed tool boundaries, small local model workers, larger model synthesis, hosted API fallback, A/B variants, tool call telemetry, cost aware routing, local LLM endpoints, and VPN secured Linux/GPU services.

Evidence Backed Research Pipeline

- Built document and source grading, evidence ledgers, critique loops, persistent research artifacts, and sample research packets for auditable research workflows.

CERTIFICATIONS AND TRAINING

CompTIA A+ | CompTIA Network+

Selected coursework: HarvardX/edX CS50AI, CS50SQL, MITx Foundations of Modern Finance, Machine Learning with Python.